



VIPEKEE Sales and Analytics Report



Tumi Sefolo

Project Overview

This project focuses on analyzing coffee shop sales data using Power BI. The dashboard was designed to provide insights into sales performance, customer purchasing behavior, beverage popularity, and peak operating hours.

The purpose of this dashboard is to help the coffee shop make data-driven decisions related to:

- Product performance
- Staffing optimization
- Peak sales periods
- Revenue generation
- Customer purchasing trends

Business Problem

VIPEKEE Coffee shops generate large volumes of transactional data daily, but without proper analysis, it becomes difficult to identify:

- Which beverages perform best
- When the busiest periods occur
- Which products generate the most revenue
- Customer purchasing patterns throughout the day
- Sales trends over time

This dashboard transforms raw transactional data into actionable insights for business decision-making.

Objectives

The objectives of this project are:

1. Analyze total sales performance
2. Identify the busiest hours of the day
3. Determine top-selling beverages
4. Analyze sales trends over time
5. Understand customer purchasing behavior
6. Monitor revenue performance by beverage and time period

Dataset Description

The dataset contains coffee shop transactional sales data.

Columns Used

Column Name	Description
SaleID	Unique transaction identifier
DateTime	Date and time of transaction
Beverage Name	Name of beverage purchased
Unit Price	Price of the beverage
Card Number	Payment card identifier

Data Preparation Process

Step 1: Import Data

The dataset was imported into Power BI using:

- Excel / CSV source
- Power Query Editor for transformation

Step 2: Data Cleaning

The following cleaning steps were performed:

- Removed null values
- Verified data types
- Standardized beverage names
- Split Date and Time into separate date and time components

Step 3: Derived Columns

Additional calculated columns were created:

Hour Label Column

Used to identify the busiest hours.

Hour Label = FORMAT(Coffee_Shop_Facts[time], "HH:00")

Used to count the number of transactions.

```
Total Transactions = COUNT(Coffee_Shop_Facts[BeverageKey])
```

Used to identify the day of the week based on transaction date.

```
Day_Type = IF(WEEKDAY('Dim_Payment'[Date.1], 2) >= 6,  
    "Weekend",  
    "Weekday")
```

Used to identify the season.

```
SeasonType = VAR MonthNum = MONTH(SELECTEDVALUE('Dim_Payment'[Month]))  
  
RETURN  
  
SWITCH( TRUE(),  
    ISBLANK(MonthNum), "Select a Date",  
    MonthNum IN {12, 1, 2}, "Summer",  
    MonthNum IN {3, 4, 5}, "Autumn",  
    MonthNum IN {6, 7, 8}, "Winter",  
    MonthNum IN {9, 10, 11}, "Spring",  
    "Unknown")
```

Data Modeling

A star schema model was implemented.

Fact Table

Fact_Sales

Contains transactional sales data.

Measures Included

- Total Revenue
 - Total Transactions
 - Average Order Value
 - Peak hour transactions
 - Season type
-

Dimension Tables

Dim_Date

Contains:

- Date
- Year
- Month
- Day

Dim_Time

Contains:

- Time
- Hour
- Time Period

Dim_Beverage

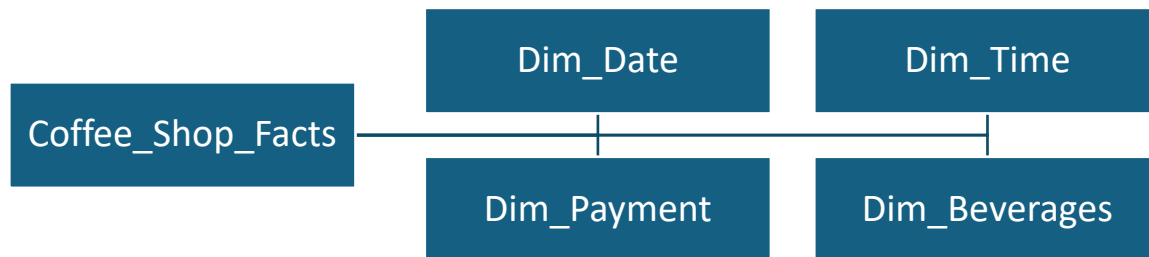
Contains:

- Beverage Name
- Card used
- Beverage cost
- Beverage size

Dim_Payment

- Hour
- Time Period
- Card used
- Payment type

Star Schema Design



Dashboard Pages

1. Executive Overview

Purpose: Provide a high-level summary of business performance.

KPIs Included

- Total Revenue
- Total Transactions
- Average Order Value
- Top Beverage

Visuals Used

- KPI Cards
- Drill down
- Revenue Trend Line Chart
- Beverage Revenue Bar Chart

2. Time Analysis

Purpose: Analyze sales performance across different times.

Key Questions Answered

- What are the busiest hours?
- Which time period generates the most revenue?
- Which beverages sell best at different times?

Visuals Used

- Sales by Hour Column Chart

- Revenue by Time Period
- Beverage Sales by Hour Stacked Chart

3. Beverage Performance

Purpose: Analyze beverage popularity and revenue contribution.

Key Questions Answered

- Which beverages generate the most revenue?
- Which beverages are most frequently purchased?
- Which beverages underperform?
- What is the average product cost?

Visuals Used

- Top Beverage Bar Chart
- Revenue Contribution Pie Chart
- Beverage Ranking Table

Key Insights

Insights included from the dashboard:

- Morning hours represent peak transaction activity.
- Latte and Cappuccino are the highest-performing beverages.
- Afternoon sales decline compared to morning peaks.
- Premium beverages contribute significantly to total revenue.
- Other methods of payment and Instant EFT rank quite high.

Business Recommendations

Based on the analysis, the following recommendations can be made:

1. Increase staffing during peak morning hours.
2. Promote slower-selling beverages using discounts.
3. Introduce afternoon promotions to increase off-peak sales.
4. Focus inventory planning around high-performing beverages.
5. Use customer purchasing patterns to optimize menu offerings.

Challenges Encountered

- Incomplete beverage categorization
- Limited customer information
- Data cleaning inconsistencies

Skills Demonstrated

This project demonstrates:

- Data cleaning in Power Query
- Data modeling using star schema
- DAX calculations
- Time intelligence analysis
- Business intelligence dashboard design
- Data storytelling
- KPI development
- Visualization best practices

Tools Used

Tool	Purpose
Power BI Desktop	Dashboard development
Power Query	Data transformation
DAX	Calculated measures and columns
Excel / CSV	Data source

Future Improvements

Potential future enhancements include:

- Adding quantity sold
- Including customer demographics
- Integrating loyalty program data
- Forecasting future sales trends
- Creating advanced customer segmentation

- Adding geographic sales analysis